HybridAD: A Hybrid Model-Driven Anomaly Detection Approach for Multivariate Time Series

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***Abstract*—Anomaly detection, in recent years, has gained increasing attention in the research and practice of time series processing. However, the task is particularly challenging with multivariate time series which complicates the temporal dependency between observations and introduces complex inter-channel correlation. Meanwhile, in order to fit a broader range of applications, robustness during both training and detection is also a critical aspect. In this paper, we propose an unsupervised, hybrid model-driven anomaly detection scheme capable of (1) transforming sequences into a fused representation of temporal dependency embeddings and inter-channel correlation embeddings, and (2) achieving robust anomaly detection using a temporal prediction network for sample-wise posterior estimation combined with a data reconstruction network to assess the source of prediction. On this basis, we develop a probability density-based anomaly scoring mechanism for online detection in multivariate time series, where the anomaly score for each observation is rectified by the reliability of the prediction source. The results of extensive experiments on five publicly available datasets show that our proposed solution outperforms various state-of-the-art anomaly detection algorithms (including DL-based and non-DL-based), achieving a performance improvement (in F1-Score) by up to 10.42%.**

***Index Terms*—Anomaly detection, multivariate time series, unsupervised learning, deep learning.**

I. Introduction

In recent years, automated anomaly detection has been widely employed in production systems concerning domains including finance, cybersecurity, and industrial process control. It also involves a broad range of critical applications such as credit card fraud detection, intrusion detection, fault diagnosis, etc [1]. The application of deep learning in anomaly detection tasks is becoming increasingly widespread [2]. Unlike other types of anomaly detection tasks, anomaly detection for time series data often requires consideration of temporal dependencies between the objects being detected [3], making it a challenging task and a significant research topic. Depending on the dimension of the data, time series can be classified as either univariate time series (UTS) or multivariate time series (MTS) [4]. In comparison to UTS, the presence of MTS extends the dimensionality and is more common in real-world systems where any critical changes are reflected by the dynamics of multiple variables that may correlate with each other. (Fig. 1 shows the trace of server performance indicators as an example). Aside from the intrinsic patterns exhibited along the temporal dimension, the interplay of multiple channels brings additional challenges to the anomaly detection for MTS especially when no prior knowledge or reliable data sources are available.

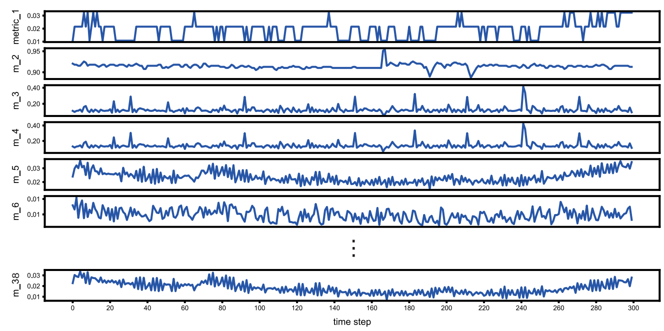

Fig. 1. Example of multivariate time series as a collection of server performance indicators (from SMD dataset [5]) that vary over 300 time steps.

Intuitively, an anomaly detection system works by distinguishing normal data from anomalous data in a similar way to the binary classification problem. However, it is usually not practical in the context of Big Data for many reasons [6]. Specifically, classification-based approaches are greatly affected by (i) severe data imbalance, which means that the proportion of anomalous data is much smaller than that of normal data in a time series; and (ii) high cost of manual labeling, which means that labeling each data point is cost-prohibitive and thus we have to deal with unlabeled data in most cases. In this regard, unsupervised methods are frequently adopted as a promising

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alternative to supervised or semi-supervised methods for anomaly detection [7], [8], [9], [10]. Despite the way of training and the model to be used, it is always essential to capture the temporal dependency between observations within different ranges of a time series. For MTS, inter-channel correlation, characterized as a complicated linear or non-linear relationship between different variables within the same time period, should also be considered particularly in anomaly detection systems [3]. However, traditional unsupervised anomaly detection methods, such as local outlier factor (LOF) [11], principal component analysis (PCA) [12], one-class SVM (OCSVM) [13], and Isolation Forest (IF) [14], are unable to extract the temporal dependency and inter-channel correlation of MTS effectively and therefore fail to meet the requirements of the majority of anomaly detection systems. DL-based approaches, such as the variants of recurrent neural networks (RNNs) and one-dimensional convolutional neural networks, have been proven effective in terms of time series feature extraction [10], [15], [16], [17]. Studies on DL-based unsupervised anomaly detection usually make the following assumptions: (i) the training data contains no or a negligible number of anomalous samples; (ii) the data patterns that anomalous data display differ significantly from those of normal data [6]. These assumptions encourage prediction-driven and reconstruction-based anomaly detection where a deep model is trained to learn normal series patterns.

Following this rationale, a number of practical DL-based algorithms have been developed for time series anomaly detection [7], [18], [19], [20], [21], [22]. However, there are still several limitations. First, existing networks often struggle in simultaneously learning the temporal dependency and inter-channel correlation of MTS. Second, the majority of existing anomaly detection models are either based on temporal prediction or data reconstruction, resulting in reduced effectiveness during training or prediction in the presence of anomalies. Finally, the performance improvement of anomaly detection model is hindered by the absence of an efficient anomaly scoring mechanism, resulting in anomalies being easily overlooked.

In this work, we propose HybridAD (**Hybrid** Model-driven **A**nomaly **D**etection Approach for MTS), a hybrid anomaly detection approach based on deep learning. For the first limitation, we design a feature extraction module mainly consisting a Gated Recurrent Unit (GRU) and a one-dimensional convolutional neural network to extract the inter-channel correlation and temporal dependency feature of MTS, respectively. Then we build a hybrid anomaly detection model that is jointly optimized based on temporal prediction and data reconstruction to address the second limitation. A novel anomaly scoring mechanism that focuses on prediction probability density is presented to enhance the anomaly detection performance.

To summarize, the main contributions of our work are as follows:

- We design a feature extraction module that combines a GRU network and a one-dimensional convolutional neural network to simultaneously learn the temporal dependency and inter-channel correlation of MTS.
- We propose a hybrid model-driven framework empowered by a temporal prediction network and a data reconstruction network for robust anomaly detection.
- We present an anomaly scoring mechanism that focuses on prediction probability density. By taking the reliability of the prediction source into account, the effectiveness of anomaly detection is further improved.
- Our experimental results on five real-world datasets show that the proposed HybridAD outperforms several state-of-the-art models for MTS anomaly detection, achieving a maximum performance improvement of 10.42% in F1-Score.

The rest of this article is organized as follows: Section II discusses the studies related to DL-based anomaly detection models. In Section III, we introduce in detail the proposed unsupervised anomaly detection scheme. In Section IV, we present and analyze the experimental results, and finally conclude the article in Section V.

II. Related Work

Due to the fact that DL-based anomaly detection models are typically trained on anomaly-free datasets to learn the normal data patterns, greater detection error will be produced in case that the anomalies exist. Based on the different techniques, DL-based anomaly detection models can be categorized as follows.

Temporal prediction-based models: On the basis of normal pattern derived from the historical data, the model could detect the anomaly through the difference between the predicted and real value of the incoming observation. RNN-based models [10], [15], [23] were proposed for temporal prediction and usually determined the anomalies utilizing the prediction errors. Specially, Hundman et al. [23] built an LSTM-based model for each channel of MTS. However, simplistically combining the anomaly detection results of multiple UTS may neglect the anomalies of inter-channel correlation. Incorporating low-dimensional embeddings to capture temporal dependencies [24], [25] and leveraging graph structures to capture inter-channel correlation in time series [21], can further improve the efficiency of anomaly detection models. However, these approaches do not simultaneously consider both types of feature factors during the time series modeling process. Meanwhile, to the best of our knowledge, there are very few studies on anomaly detection that consider the reliability of the prediction source. Similar to [10], our work also takes into account the reliability of the prediction source.

Data reconstruction-based models: When an input sequence deviates from the normal pattern due to the existence of anomalies, a well-trained reconstruction model will struggle to recover the input sequence and output greater reconstruction errors. The Variational Auto-Encoders (VAE)-based models [22], [26], [27] were proposed to reconstruct the sequences' expected distribution and used the reconstruction probability as anomaly score. Zong et al. [20] used a deep autoencoder to generate a low-dimensional representations from both the reduced space and the reconstruction error features, which were then fed to a Gaussian mixture model to estimate their likelihood. However, the robustness of the model in anomaly detection using only data reconstruction needs further improvement. Some works conducted an anomaly detection scheme based on the paradigm of hybrid models [6], [19], but the error-based methods for

TABLE I
LIST OF SYMBOLS

symbol	description
W	The set of sequences obtained after window slicing, each one of size L.
W_t	The sequence in the time interval $[t-L+1, t]$ $(t \geq L)$.
$\hat{W}_t$	The reconstructed sequence in the time interval $[t-L+1, t]$ $(t \geq L)$.
x	The observation in sequence.
x_t	The observation at the moment t in sequence, represented by a vector.
x_t^m	The m^{th} feature (or channel referred to [10]) of the observation at the moment t in sequence, represented by a scalar.
$\hat{x}_t^m$	The m^{th} feature of the observation predicted at the moment t in the sequence.
z	The latent variable in VAE model.
z_t	The latent variable representing the input sequence W_t.
z_t^i	The i^{th} dimension of the latent variable z_t.

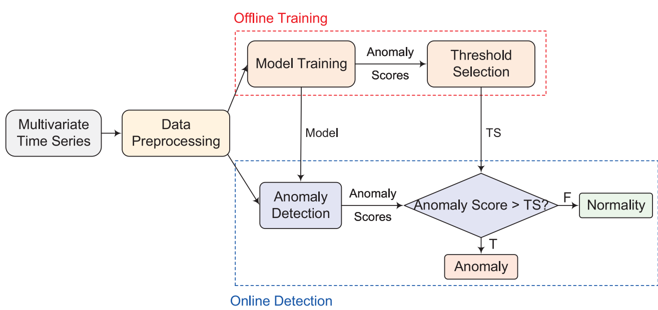

Fig. 2. Overall process of unsupervised anomaly detection for MTS. It consists of an offline training phase and an online detection phase.

anomaly detection faced challenges in distinguishing anomalies of different scales. Inspired by GAN [28] that enables the model to fit the target distribution of any dataset through adversarial training, [18], [29], [30] proposed a GAN-based model to perform anomaly detection for time series. However, the instability of the adversarial training may hinder the efficiency and effectiveness of anomaly detection in practice.

III. METHODOLOGY

In this section, we present the overall process of unsupervised anomaly detection for MTS and introduce the design of the proposed anomaly detection model HybridAD in detail. In addition, we also provide an anomaly scoring mechanism and a threshold selection strategy, which make significant influence on model's performance. For clarity, Table I lists all the symbols frequently used in this article.

A. Overview

The workflow of our anomaly detection system involves two phases: **offline model training** and **online anomaly detection** (as depicted in Fig. 2).

Offline model training: During this phase, the data preprocessing module receives a MTS and outputs a set of sequences (after window slicing and data normalization), each of which is formatted as $W_t = \{x_{t-L+1}, \ldots, x_t\} (t \geq L)$. The sequences are loaded into the model training module (which details in Section III-B), allowing the model to learn the temporal dependency and inter-channel correlation features of MTS in a normal pattern. After training, the model will provide anomaly score for each observation in the training set and employ an adaptive threshold selection strategy to generate the anomaly threshold *TS* for the online anomaly detection.

Online anomaly detection: After training, the model is able to detect the MTS input in real-time and assign a score to each observation. Observations with scores over the threshold determined in offline training are considered anomalous.

B. Design of HybridAD

As shown in Fig. 3, this article proposes a probability density-based anomaly detection model combined with temporal prediction and data reconstruction to address the challenges in MTS anomaly detection. The data pre-processing module divides the MTS into a set of sequences based on the specified window size. The feature extraction network permits the efficient extraction of temporal dependency and inter-channel correlation. Finally, the joint optimization of the anomaly detection model is achieved by feeding the fused embeddings to the data reconstruction network and the temporal prediction network, respectively.

Feature Extraction Network: To extract the inter-channel correlation and temporal dependency of MTS simultaneously, the feature extraction network employs a GRU module and a one-dimensional convolutional neural module. In particular, the GRU module is used to obtain inter-channel embedding, and the length L of the compressed series remains constant while the number of dimensions decreases to $M'(M' < M)$. The one-dimensional convolutional module is used to obtain temporal embedding with a fixed number of dimensions, and the length of the series decreases to $L'(L' < L)$. With the inter-channel embedding h_{ch} and temporal embedding h_{tp} as input to the fully connected layer, the embedding fusion is performed. Then the fused embedding h_t is fed to the temporal prediction network and data reconstruction network to achieve joint optimization.

Temporal Prediction Network: One of the typical ways used in temporal prediction-based anomaly detection model updating is based on the prediction error. In contrast to the majority of temporal prediction-based anomaly detection models, HybridAD employs maximum likelihood estimation to fit the probability distribution of future observations, and the model's output is the probability density of the observation under that distribution. Similar to other works on probability prediction by assuming an underlying distribution of the time series [31], [32], in our work, we assume that each feature of the future observation $x_{t+1} = [x_{t+1}^1, x_{t+1}^2, \ldots, x_{t+1}^M]$ follows a Gaussian distribution $N(\mu_{x_{t+1}^i}, \sigma_{x_{t+1}^i})$, where i represents the i^{th} feature of x_{t+1} and $i \leq M$. The loss function of the temporal prediction-based model is defined as the negative logarithm of the likelihood function:

$$\text{Loss}_{pre} = -\log \prod_{i=0}^{M} p\left(x_{t+1}^i \mid \mu_{x_{t+1}^i}, \sigma_{x_{t+1}^i}\right), \qquad (1)$$

where $\mu_{x_{t+1}^i}$ and $\sigma_{x_{t+1}^i}$ are the mean and standard deviation of the distribution, obtained from a liner layer and a Softplus layer, respectively.

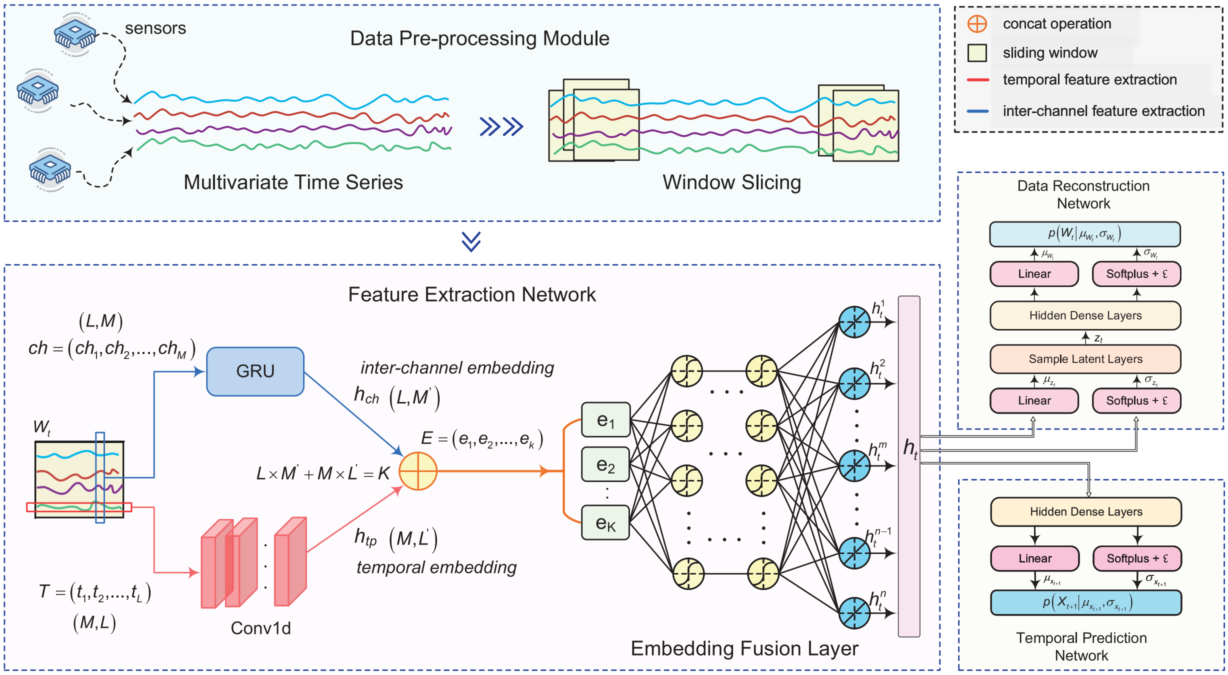

Fig. 3. Design of hybrid anomaly detection model.

Data reconstruction network: The data reconstruction network is designed based on the idea of Variational Autoencoder [33]. In this work, the feature extraction network and data reconstruction network are combined to form a complete variational autoencoder structure. The main component of the data reconstruction network is a multilayer, fully-connected neural network. Assume the prior $p_\theta(z_t)$ follows a normal distribution $N(0, I)$, where z_t is a latent representation with a reduced dimension in VAE, representing a high-dimensional input W_t. The posterior distributions of W_t and z_t are described by the diagonal Gaussian distributions $p_\theta(W_t \mid z_t) = N(\mu_{W_t}, \sigma^2_{W_t} I)$ and $q_\phi(z_t \mid W_t) = N(\mu_{z_t}, \sigma^2_{z_t} I)$, respectively, where the parameters μ and σ^2 of the Gaussian diagonal distributions are obtained from a linear layer and a Softplus layer, respectively. The Softplus layer ensures that the standard deviation σ of the output is consistently greater than 0. A constant ϵ is added to the Softplus layer to address the issue that the parameter σ may be too small for training the model [26]. As input to the decoder, the hidden state distribution is sampled to produce the vector z_t, which is then used to reconstruct the distribution of the sequence W_t. It should be noted that the output of the VAE in this work is not the reconstructed sequence but rather the mean and standard deviation of the distribution that the reconstructed sequence follows. The loss function of the data reconstruction network can be defined as:

$$\begin{aligned}\text{Loss}_{rec} = & KL\left(q_\phi\left(z_t \mid W_t\right) \| p_\theta\left(z_t\right)\right) \\ & + E_{z_t \sim q_\phi}\left[\log p_\theta\left(W_t \mid z_t\right)\right], \end{aligned} \tag{2}$$

where the first term represents the degree of similarity between the approximate posterior distribution $q_\phi(z_t \mid W_t)$ of the latent variable z_t and the prior distribution $p_\theta(z_t)$, which can be calculated by using the KL divergence. Under the assumption that the prior distribution of the latent variable z_t is a standard Gaussian distribution, the KL divergence of the two distributions above is calculated as (where ‘ $\sim$ ‘ denotes $q_\phi(z_t \mid W_t) \| p_\theta(z_t)$):

$$KL\left(\sim\right) = -\frac{1}{2}\sum_{i=0}^{D}\left(1 + 2\log\sigma_{z_t^i} - \mu^2_{z_t^i} - \sigma^2_{z_t^i}\right), \tag{3}$$

where D is the size of the latent variable z_t. The second term in (2) represents the degree of similarity between the given input W_t and the reconstructed sequence, which can be calculated using the maximum likelihood estimation as follows (where ‘ $\sim$ ‘ denotes $\log p_\theta(W_t \mid z_t)$):

$$E_{z_t \sim q_\phi}\left[\sim\right] = \frac{1}{L}\sum_{i=t-L}^{t}\left(-\log\prod_{j=0}^{M} p\left(x_i^j \mid \mu_{x_i^j}, \sigma_{x_i^j}\right)\right). \tag{4}$$

Based on the parameters output by the data reconstruction network, the reconstruction probability density of the input sample W_t can be inferred. W_t is used as a prediction source for future observation x_{t+1} in the temporal prediction network. In this work, the reconstruction probability density of the input W_t is utilized to assess the reliability of the prediction source. When the prediction source W_t contains anomalies (i.e., the reconstruction probability density is below a normal level), the prediction result x_{t+1} based on the input W_t will be considered unreliable. The anomaly score (to be introduced in Section II-I-C) can be amplified further by integrating the reliability of the prediction source, thereby making the distinction between anomalous and normal data more obvious and improving the accuracy of anomaly detection.

By the joint optimization by the temporal prediction and data reconstruction network, the loss of the hybrid anomaly detection

Algorithm 1: HybridAD Training Algorithm.

Input: training set $X_{\text{train}} = \{(W_t, x_{t+1})\}(t \geq L)$, iteration epoch number N, feature extraction network $fNet$, temporal prediction network $pNet$, data reconstruction network $reNet$.
Output: trained $fNet, pNet, reNet$
1: $fNet, pNet, reNet \leftarrow$ initialize parameters
2: $n \leftarrow 1$
3: **repeat**
4: **for** (W_t, x_{t+1}) in X_{train} **do**
5: $h_t \leftarrow fNet(W_t)$
6: $\mu_{x_{t+1}}, \sigma_{x_{t+1}} \leftarrow pNet(h_t)$
7: $z_t, \mu_{W_t}, \sigma_{W_t} \leftarrow reNet(h_t)$
8: compute Loss_{pre} via (1)
9: compute Loss_{rec} via (2)
10: $\text{Loss} \leftarrow \text{Loss}_{pre} + \text{Loss}_{rec}$
11: $fNet, pNet, reNet \leftarrow$ model parameters optimization according to the Loss
12: **end for**
13: $n \leftarrow n + 1$
14: **until** $n = N$
15: **return** trained $fNet, pNet, reNet$

model can be calculated as:

$$\text{Loss} = \text{Loss}_{pre} + \text{Loss}_{rec}. \tag{5}$$

The training process of HybridAD model can be described in Algorithm 1.

C. Anomaly Scoring

On the basis of HybridAD, we propose an anomaly scoring mechanism as a mixture of the prediction probability density and reconstruction probability density.

Prediction probability density: For the sequence $W_t = \{x_{t-L+1}, \ldots, x_t\}$ with a given window size L, the temporal prediction network fits the probability distribution $p(x_{t+1} \mid W_t)$ of the future observation x_{t+1} via maximum likelihood estimation. The anomaly score is then calculated as the probability density of the observation under the distribution $p(x_{t+1} \mid W_t)$. The lower the probability density, the more likely the observation is an anomaly. Consequently, the score of the observations for the temporal prediction model can be calculated as:

$$\begin{aligned} p_p &= p\left(x_{t+1} \mid \mu_{x_{t+1}}, \sigma_{x_{t+1}}\right) \\ &= \prod_{i=0}^{M} \left(p\left(x_{t+1}^i \mid \mu_{x_{t+1}^i}, \sigma_{x_{t+1}^i}\right)\right). \end{aligned} \tag{6}$$

Reconstruction probability density: For the sequence $W_t = \{x_{t-L+1}, \ldots, x_t\}$, the data reconstruction network outputs the mean and standard deviation of the distribution $p_\theta(W_t \mid z_t)$ of the reconstructed sequence. Therefore, the reconstruction probability refers to the probability density of the sequence W_t under the distribution $p_\theta(W_t \mid z_t)$. In the temporal prediction network, the reliability of the prediction result for x_{t+1} is significantly dependent on the proportion of normal data points in the sequence W_t (as the prediction source). As a result, we rectify the anomaly score by using the reconstruction probability to measure the reliability of the prediction source. Moreover, we believe that data closer to the predicted point x_{t+1} is more crucial, so we employ an exponentially decaying weights approach that assigns a weight $d_i = \alpha^{i-(t-L+1)}$ to each observation in sequence W_t. The final reconstruction probability after weighting rectification is:

$$p_r = \frac{\prod_{i=t-L+1}^{t} \left(d_i \times \prod_{j=0}^{M} p\left(x_i^j \mid \mu_{x_i^j}, \sigma_{x_i^j}\right)\right)}{D_L}, \tag{7}$$

where $D_L = \sum_{i=t-L+1}^{t} d_i$.

In summary, this work provides an anomaly score mechanism that focuses on prediction probability density while also incorporating the reliability of the prediction source, which allows the model to distinguish normal data from anomalous data more accurately. The anomaly score $S_{x_{t+1}}$ for the observation x_{t+1} is defined in this study as the negative logarithm of the product of the prediction source's reconstruction probability density and the prediction probability density, as given in (8).

$$S_{x_{t+1}} = -\log\left(p_p \times p_r\right). \tag{8}$$

D. Threshold Selection Strategy

In practice, using the manual selection of score threshold in complex anomaly detection systems not only requires expert knowledge but also compromises the robustness of the solution. Extreme Value Theory (EVT), a statistical theory that explores the extreme value law, has been applied in some anomaly detection work [12], [34]. Peaks Over Threshold (POT) is the second theory of EVT, which is used to fit the tail distribution of the data probability distribution. This work uses a POT method to concentrate on the high end of the distribution which the extreme values (i.e., the scores of anomalies) in the set of scores follows [5], [19]. Algorithm 2 depicts the anomaly detection flow of the HybridAD model combined with the POT-based threshold selection strategy. Note that applying POT to achieve anomaly threshold selection is not the main contribution in this work.

IV. Experiments and Evaluation

In this section, we first introduce the datasets and the evaluation metric used in our experiment (Section IV-A). We compare the performance of HybridAD to that of state-of-the-art anomaly detection algorithms using the F1-Score on five publicly available datasets (Section IV-B). Then, the effectiveness of the feature extraction network and the probability density-based model are validated through a series of ablation experiments (Section IV-C). Finally, we discuss the effect of the POT-based threshold selection strategy and the hyperparameter on model's performance (Section IV-D).

TABLE II
OVERVIEW OF THE FIVE PUBLICLY AVAILABLE DATASETS

dataset	size of training set	size of test set	number of dimensions	ratio of anomalies (%)
SWaT	99360	89984	51	11.98
WADI	241921	15701	127	7.09
SMD	708405	708420	38	4.16
SMAP	135183	427617	25	13.13
MSL	58317	73729	55	10.72

Algorithm 2: Anomaly Detection Algorithm Based on HybridAD.

Input: window length L, trained $fNet$, $pNet$ and $reNet$, hyperparameters q and $level$ associated with POT, training set $X_{\text{train}} = \{(W_t, x_{t+1})\}(t \geq L)$, test set $X_{\text{test}} = \{(W'_t, x'_{t+1})\}(t \geq L)$.
Output: anomaly detection results $Y = \{y_{t+1}\}(t \geq L)$

```
S_train ← ∅
for (W_t, x_{t+1}) in X_train do
    h_t ← cNet(W_t)
    μ_{x_{t+1}}, σ_{x_{t+1}} ← pNet(h_t)
    z, μ_{W_t}, σ_{W_t} ← reNet(h_t)
    compute S_{x_{t+1}} via (8)
    add S_{x_{t+1}} to S_train
end for
th ← POT_threshold_selection(S_train, q, level)
Y ← ∅
for (W'_t, x'_{t+1}) in X_test do
    h'_t ← fNet(W'_t)
    μ_{x'_{t+1}}, σ_{x'_{t+1}} ← pNet(h'_t)
    z'_t, μ_{W'_t}, σ_{W'_t} ← reNet(h'_t)
    compute S_{x'_{t+1}} via (8)
    if S_{x'_{t+1}} > th then
        y'_{t+1} ← 1
    else
        y'_{t+1} ← 0
    end if
    add y'_{t+1} to Y
end for
return Y
```

A. Datasets and Evaluation Metric

In our experiments, the performance of the HybridAD model is evaluated on five publicly available datasets: *Secure Water Treatment (SWaT)* [35], *Water Distribution (WADI)* [29], *Server Machine Dataset (SMD)* [5], *Soil Moisture Active Passive (SMAP) satellite* and *Mars Science Laboratory (MSL) rover* [23]. The overview of each dataset is shown in Table II. To demonstrate the performance of the models, we mainly use F1-Score. In practice, anomalies in time series are typically exhibited as successive segments of anomalous data rather than as a single anomaly. Similar to [26], we consider an anomalous sequence to be correctly detected if the model detects at least one anomalous observation in the sequence, which implies that all other anomalies in the anomalous sequence are also deemed to be correctly detected. More details are in Appendix A.

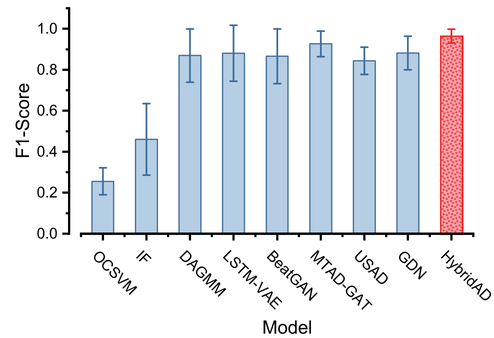

Fig. 4. Performance (with standard deviation) of HybridAD and other baseline methods measured in F1-Score.

B. Performance Analysis

In this experiment, we compare HybridAD's performance with both DL-based and non-DL-based methods, including OCSVM [13], Isolation Forest (IF) [14], DAGMM [20], LSTM-VAE [22], BeatGAN [18], MTAD-GAT [19], USAD [7], GDN [21] algorithms, and adopt the optimal threshold for each method when calculating the corresponding F1-Score. Table III shows the performance of all models on five datasets. Note that the symbol **P**, **R** and **F1** in Table III denote precision, recall, and F1-Score, respectively, where the **bold** denotes the best performance and the *italic* denotes the second-best performance. It can be seen that the proposed hybrid model-driven method achieves the best performance on all the datasets. In addition, we find that the majority of deep learning-based models (other than OCSVM and IF) performed well on SMD, SMAP, and MSL, as the anomaly patterns of the sequences in these three datasets are relatively simple, consisting primarily of temporal dependency anomalies with substantial numerical variations in some of dimensions. Even when the anomaly pattern is straightforward, HybridAD outperforms the other models in terms of F1-Score. For SWaT and WADI with more complex anomaly patterns, the performance advantage of HybridAD is more significant. Especially on the WADI, only MTAD-GAT obtains better performance among all baseline models with F1-Score of 0.8811, while HybridAD achieves a 10.42% performance improvement with an F1-Score of 0.9729. In Appendix B-D, we further provide significance test results for reference. As shown in Fig. 4, when the performance of all models is averaged across all datasets, HybridAD achieves the highest average performance and more excellent stability.

C. Ablation Study

1) Effectiveness of the Feature Extraction Network: Effective extraction of feature information from time series is of great importance for improving the performance of the model. In this article, we designed a feature extraction network composed of a GRU network and a one-dimensional convolutional network to extract the inter-channel correlation and temporal dependency information of the time series. To verify the effectiveness of these components, we designed three model variants. In the first variant (**HybridAD_Wo_T**), we removed the one-dimensional

TABLE III
PERFORMANCE OF EACH MODEL ON THE PUBLICLY AVAILABLE DATASETS

Methods	SWaT			WADI			SMD			SMAP			MSL		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
OCSVM	0.1355	0.9884	0.2383	0.1765	1.0000	0.3000	0.0815	0.9993	0.1507	0.1894	1.0000	0.3185	0.1564	1.0000	0.2704
IF	0.1566	0.9673	0.2695	0.3448	0.7718	0.4767	0.1775	0.9905	0.3011	0.5545	0.8412	0.6684	0.4270	0.9470	0.5886
DAGMM	0.9298	0.7685	0.8415	0.7178	0.6056	0.6569	0.9285	0.9791	0.9531	0.9644	0.9985	0.9811	0.8417	1.0000	0.9141
LSTM-VAE	0.9875	0.6984	0.8182	0.5400	0.8787	0.6689	0.9481	0.9789	0.9633	0.9725	0.9981	*0.9851*	0.9409	0.9974	0.9683
BeatGAN	0.9416	0.7237	0.8184	0.9423	0.4987	0.6522	0.9151	0.9621	0.9381	0.9189	0.9970	0.9563	0.9638	0.9670	0.9654
MTAD-GAT	0.9311	0.7701	0.8429	0.8835	0.8787	*0.8811*	0.9644	0.9959	*0.9799*	0.8966	0.9995	0.9453	0.9687	0.9955	*0.9819*
USAD	0.9025	0.7872	0.8409	0.6581	0.8787	0.7526	0.9338	0.9422	0.9380	0.7101	0.9961	0.8291	0.8032	0.9254	0.8600
GDN	0.9532	0.7988	*0.8691*	0.9750	0.6056	0.7471	0.9579	0.9106	0.9337	0.9253	0.9887	0.9559	0.8761	0.9275	0.9011
HybridAD	0.9433	0.8878	**0.9147**	0.9472	1.0000	**0.9729**	0.9904	0.9699	**0.9801**	0.9804	0.9987	**0.9894**	0.9689	1.0000	**0.9842**

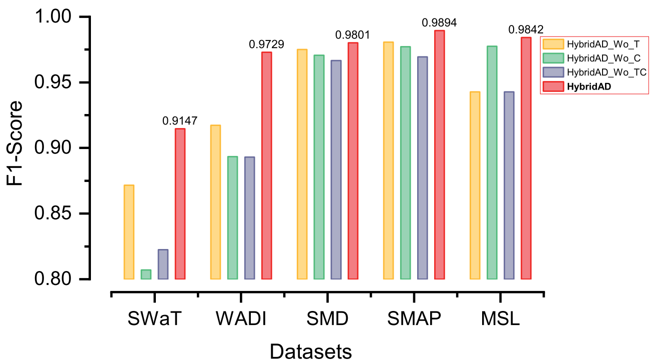

Fig. 5. Results of ablation experiments on the effectiveness of the feature extraction network.

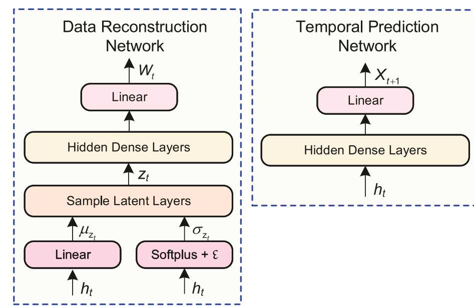

Fig. 6. Variant models in ablation studies. The variant model of the network for data reconstruction is shown on the left, and the variant model of the network for temporal prediction is shown on the right.

convolutional network to make the module not explicitly extract temporal dependency information. In the second variant (**HybridAD_Wo_C**), we removed the GRU network to make the module not explicitly extract channel correlation information. In the third variant (**HybridAD_Wo_TC**), we removed the entire feature extraction module and directly input the preprocessed sequence into the prediction network and reconstruction network. Fig. 5 shows the results of the experiment.

The results in Fig. 5 indicate that in time series anomaly detection tasks, extracting temporal features and using them as inputs to downstream models (which are temporal prediction and data reconstruction networks in this article) is effective for improving model performance. As temporal dependency and inter-channel correlation are important features of MTS, considering a single feature in the model cannot achieve optimal performance.

2) Effectiveness of the Probability Density-Based Model: To validate the effectiveness of probability density-based model in HybridAD, the probability-based prediction model is replaced by a value-based prediction model, and its loss function is modified to (9). As shown in (10), the likelihood of the reconstructed sequence in the data reconstruction model is modified to be accounted for by the mean square error. The structure of the modified model variants of data reconstruction network and temporal prediction network is given in Fig. 6.

$$\text{Loss}_{pre} = \left\| x_{t+1} - \hat{x}_{t+1} \right\|_2^2, \tag{9}$$

$$\text{Loss}_{rec} = \left\| W_t - \hat{W}_t \right\|_2^2, \tag{10}$$

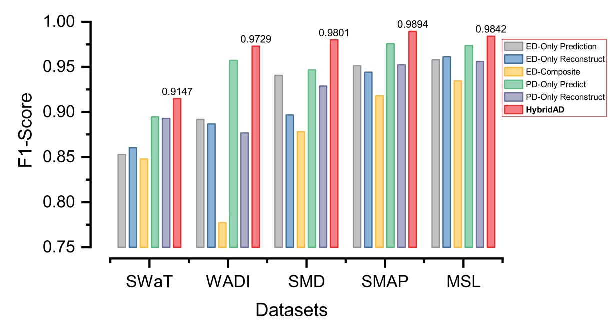

Fig. 7. Results of ablation experiments on the effectiveness of the probability density-based model.

Following the modified model variants, the anomaly scoring is calculated in accordance with the Euclidean distance (ED). This experiment takes into account only the prediction error (**ED-Only Prediction**), the reconstruction error (**ED-Only Reconstruct**), and the combination of both in anomaly scoring mechanism (**ED-Composite**), respectively. Meanwhile, the experiments additionally consider only the prediction probability density (**PD-Only Prediction**) and the reconstruction probability density (**PD-Only Reconstruct**), respectively. More details will be presented in Appendix B-E.

As depicted in Fig. 7, the anomaly detection model proposed in this article achieves superior performance on each dataset when compared to any model variant. We discover that the detection performance of the probability density-based temporal prediction model (PD-Only Prediction) is improved on the five datasets when the prediction source reliability is taken into account. The ED-Composite model variant still performs poorly,

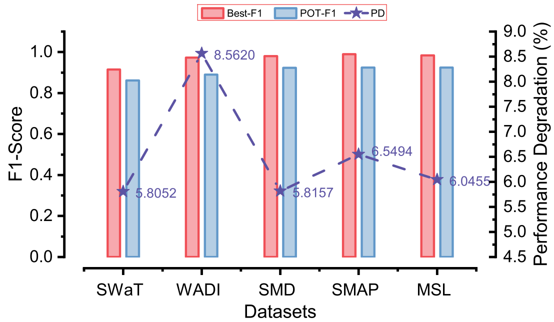

Fig. 8. Comparison of the performance of HybridAD using the optimal threshold and the POT-based threshold, respectively. 'PD' specifies the percentage of performance degradation.

even though taking the reliability of the prediction source into account. It can be observed that utilizing the probability-based anomaly scoring mechanism is superior to using the error-based one in the majority of cases, and the combination of the reliability of the prediction source can improve the detection performance of the model to some extent.

D. Parameter Study

1) Performance Degradation With POT: Without labels, it is theoretically challenging for the anomaly detection model to obtain the anomaly threshold when the optimal F1-Score is achieved. As introduced in Section III-D, we apply the POT method to HybridAD to implement the automatic selection of anomaly thresholds. Fig. 8 illustrates the performance of HybridAD after applying the POT-based threshold selection strategy. It can be seen that the performance of the model on various datasets decreases in comparison to its theoretically optimal F1-Score. Nevertheless, the performance degradation of the model is within a tolerable range ($5.81\% \sim 8.56\%$), indicating that HybridAD still achieves a superior performance that is acceptable in practical anomaly detection systems.

2) Hyperparameter Sensitivity: We study the sensitivity of our HybridAD with respect to two hyperparameters, namely the number of iteration epochs during training and the window size set during the pre-processing for MTS. Note that the hyperparameter sensitivity analysis experiments for HybridAD in this article are conducted on SWaT and WADI. On the basis of the significant performance differences among the models on SWaT and WADI in Table III, we believe that the hyperparameter sensitivity analysis experiments on these two datasets can reflect the worst performance of the HybridAD on hyperparameter sensitivity to some extent and more closely match the performance in complex anomaly detection systems.

Multiple Iteration Epochs: During model training, the training cost increases as the greater iteration epochs required to improve the performance of the model. This article focuses on whether or not HybridAD can achieve satisfactory performance and stability with fewer iteration epochs.

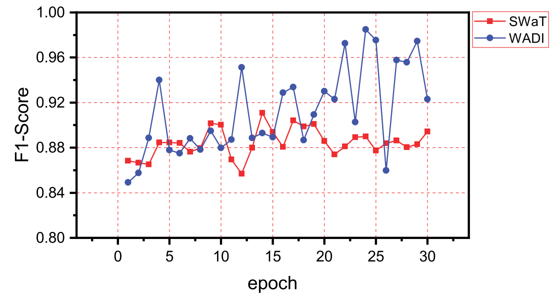

Fig. 9. Sensitivity analysis of HybridAD to the number of iteration epochs.

To explore the performance of HybridAD model under different training iteration epochs, we evaluate the model performance using F1-Score. Note that this experiment is conducted under a window size of 30. Fig. 9 demonstrates that the HybridAD model performs well on both datasets within a limited number of iteration epochs, which is indicative of the inexpensive training cost of the model. However, the performance of HybridAD on WADI fluctuates dramatically more than on SWaT. We analyze that this is due to the WADI's more complex inter-channel features (which we know has the highest number of dimensions among the five publicly available datasets with 127 according to Table II). Consequently, improving the performance stability of the model on datasets with complex inter-channel features is one of the objectives in our future work.

Sliding Window Length: One of the most frequently addressed issues in time series research is how to select an appropriate sliding window size under keeping the balance between model performance and training cost. We examine the performance (i.e., precision, recall and F1-Score) of the HybridAD model with different sliding window size settings. For each window size setting, we demonstrate the performance of HybridAD across 30 iteration epochs.

As depicted in Fig. 10, the HybridAD model performs well on both SWaT and WADI despite varying window size settings. Similar to the results of the training epoch sensitivity analysis experiments, the performance on WADI exhibits greater variation than on SWaT. Specifically, HybridAD performs better on WADI than on SWaT in terms of average recall but is less impressive in terms of average precision and F1-Score. This indicates that HybridAD has a significant capacity for learning complex anomaly patterns of MTS and identifying anomalies with a higher recall. It is a satisfactory outcome for some detection cases where missing anomalies are prohibited. However, the existence of the complex time series anomaly pattern may cause the model to incorrectly classify the normal data. Overall, the performance of the HybridAD model is still acceptable even though performance fluctuation exists. Due to the fact that real-world anomaly detection systems place a greater emphasis on model recall, the recommended settings for the sliding window on SWaT and WADI are 60 and 30, respectively, considering multiple performance metrics including recall. Although the performance of the proposed HybridAD on SWaT and WADI is

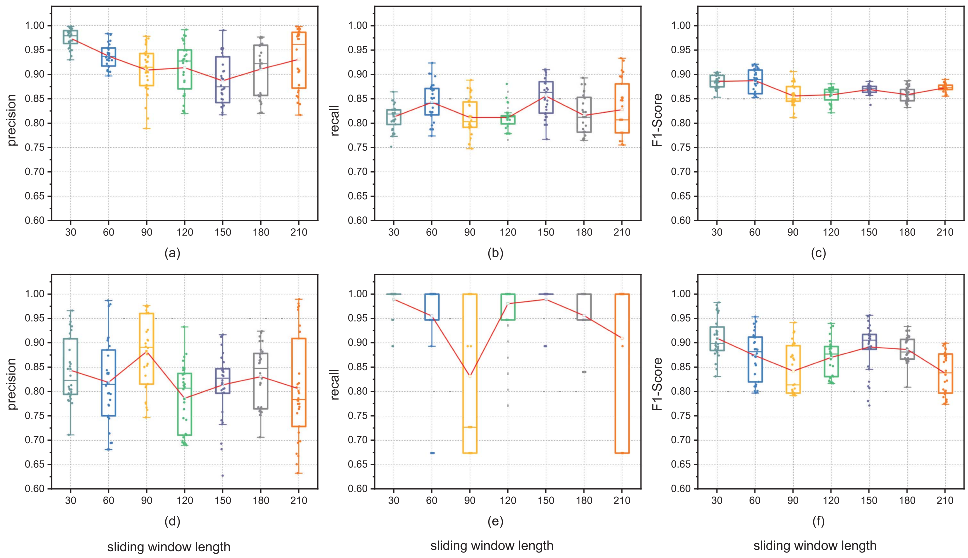

Fig. 10. Performance of HybridAD with varying sliding window sizes. Specifically, (a)∼(c) and (d)∼(f) denote the precision, recall and F1-Score of HybridAD on SWaT and WADI, respectively, with different sliding window sizes.

comparable to that of the model in the article [29], our model still achieves an improvement in terms of precision, recall, and F1-Score.

V. Conclusion

In this article, we propose an unsupervised, hybrid model-driven anomaly detection scheme targeting at complex multivariate time series. First, a feature extraction module that employs a GRU network and a one-dimensional convolutional neural network is designed to extract the inter-channel correlation and temporal dependency of multivariate time series for enhanced sequence embedding. To improve the robustness of the model, we propose a hybrid anomaly detection model that is jointly optimized by learning the posterior probability distribution of incoming observations and the probability distribution of the input sequences. In addition, an anomaly scoring mechanism focused on prediction probability density takes into account prediction source's reliability which is calculated as reconstruction probability density, thereby enhancing the anomaly detection performance. We evaluate our proposed anomaly detection algorithm on five publicly available datasets, and the experimental results show that our scheme outperforms the baseline models chosen in this article in terms of F1-Score, with a maximum performance improvement of 10.42%. We also empirically demonstrate that HybridAD only experiences minor performance loss given a sub-optimal threshold and that our model still provides satisfactory performance given smaller training budgets that concern the number of training epochs and the input sequence length.

Appendix A
Details of Datasets and Evaluation Metrics

SWaT [35] records a total of 11 days of operational data from the industrial water treatment plant, with the first 7 days in normal operation mode (i.e., the training set) and the last 4 days in an attack scenario (i.e., the test set), containing anomaly labels. Details obtained from https://itrust.sutd.edu.sg/testbeds/secure-water-treatment-swat/.

WADI [29], as an extended dataset of *SWaT*, records a total of 16 days of operation data, with the first 14 days for normal operation and the last two days for abnormal operation under the attack scenarios. Details obtained from https://itrust.sutd.edu.sg/testbeds/water-distribution-wadi/.

SMD [5] is a server machine dataset that records monitoring data for 28 servers with a total of 33 metrics over the course of 5 weeks. Details obtained from https://github.com/NetManAIOps/OmniAnomaly.

SMAP and *MSL* [23] are both expert labeled datasets from NASA containing data for 55 and 27 entities, with 25 monitored metrics per entity for *SMAP* and 55 monitored metrics per entity for *MSL*. Details obtained from https://github.com/khundman/telemanom.

To evaluate the performance of the models, the precision, recall, and F1-Score (F1 for short) are utilized and can be calculated as follows:

$$\text{precision} = \frac{TP}{TP + FP}, \tag{11a}$$

$$\text{recall} = \frac{TP}{TP + FN}, \tag{11b}$$

$$F_1 = \frac{2 \times precision \times recall}{precision + recall}, \tag{11c}$$

where TP denotes the number of correctly detected anomalous data, and FP denotes the number of normal data identified as anomalous, and FN denotes the number of anomalous data identified as normal. In practice, anomalies in time series are typically exhibited as successive segments of anomalous data rather than as a single anomaly. The schematic diagram of prediction adjustment strategy mentioned in Section IV-A is depicted in Fig. 11.

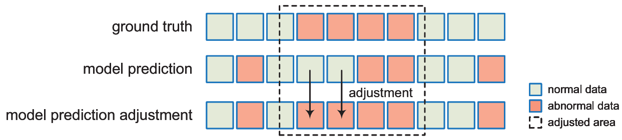

Fig. 11. Demonstration of prediction adjustment strategy.

TABLE IV
MODEL PARAMETERS TABLE

dataset	Conv1d (Kernel,Stride)	inter-channel embedding size	VAE latent variable size	POT parameters	
				q	level
SWaT	(5,2),(5,1),(5,2)	8	50	0.001	0.999
WADI	(5,2),(5,1),(5,2)	16	50	0.001	0.999
SMD	(5,2),(5,1),(5,2)	8	50	0.001	0.999
SMAP	(3,2),(3,1),(3,1)	5	30	0.001	0.99
MSL	(3,2),(3,1),(3,1)	8	30	0.001	0.99

TABLE V
TRAINING PARAMETERS TABLE

dataset	batch size	window length
SWaT	2048	30
WADI	4096	30
SMD	1024	30
SMAP	512	15
MSL	512	15

APPENDIX B
DETAILS OF EXPERIMENTS

A. Experimental Setup

Our experiments are conducted on a machine equipped with a 16-cores CPU (model: Intel(R) Xeon(R) Gold 5218 CPU @ 2.30 GHz), a GPU (model: Nvidia Tesla T4), and 256 GB of memory. The HybridAD model is implemented on the platform with Python 3.9.7 and Pytorch 1.10. Tables IV and V list the model parameters and training parameters, respectively. Specially in Table IV, the *Conv1d(Kernel, Stride)* denotes the parameter of the one-dimensional convolutional neural network. The *inter-channel embedding size* denotes the size of compressed dimensions M'. The *VAE latent variable size* denotes the size of the latent variable z in the VAE model. The *q* and *level* are hyperparameters associated with POT. *q* denotes the expected probability that the anomaly score exceeds the initial threshold. The *level* denotes the quantile, while $(1 - level) \times N_{train}$ represents the number of samples with scores exceeding the initial threshold.

Additional configurations are explained as follows. The number of layers of the one-dimensional convolutional neural network used to obtain the temporal embedding is 3. The activation functions are all designed as ReLU functions, and the Batch Normalization module is added to stabilize the training of the model. The inter-channel embedding is obtained using a GRU network of 1 layer with 128 hidden units. The body of both temporal prediction network and the data reconstruction network (i.e., the hidden dense layer) is a three-layer fully connected neural network with the structure (512, 256, 128), and the activation function is ReLU. The Batch Normalization module is also implemented. The parameter ϵ is set to 0.001 in the Softplus layer. For the training parameters not listed in Table V, the total number of epochs for model training is set to 30, and the prediction step size is set to 1. In addition to the above parameters, the parameter optimizer of the model is Adam on all datasets, and the learning rate is set to 0.001. The validation sets of SWaT, WADI, and SMD account for 10% of each dataset, and for SMAP and MSL, where the number of data samples is relatively small, the percentage is set to 30%.

TABLE VI
RESOURCE COST OF HYBRIDAD

	numbers of blueparameters	GPU memory usage (MiB)	training time per epoch (s)	testing time per point (s)
SWaT	2431300	1844 / 15109	5.7816	0.0478
WADI	5742676	7210 / 15109	19.952	0.0497
SMD	1951054	1566 / 15109	0.8164	0.0166
SMAP	903571	1362 / 15109	0.0993	0.0102
MSL	1545568	1376 / 15109	0.0933	0.0203

B. Resource Cost

In order to better demonstrate the resource costs of HybridAD in practical applications, we recorded the model parameter size, time cost per training epoch, time cost for outputting anomaly scores for each observation point, and GPU memory usage of HybridAD on five datasets, as shown in Table VI. The number of model parameters is related to the size of the input window and the number of channels. A larger window size and more channels will result in a larger number of model parameters. GPU memory usage is related to the batch size setting. A larger batch size will result in larger memory usage. All of the above factors will lead to longer training time for each epoch of the model. Nevertheless, according to Section IV-D2, we know that HybridAD can achieve good detection performance in a few training iterations, so the training time cost of HybridAD on these five datasets is acceptable. At the same time, HybridAD requires very little time to score each observation point, enabling it to detect more data points within a given time frame.

C. Baseline Algorithms Implementation

OCSVM [13] and Isolation Forest [14] use the existing implementation of scikit-learn. MTAD-GAT [19] comes from a Github implementation on https://github.com/ML4ITS/mtad-gat-pytorch. USAD [7] comes from a Github implementation on https://github.com/manigalati/usad. GDN [21] comes from the authors' implementation on https://github.com/d-ailin/GDN. The rest of the DL-based learning models [18], [20], [22] are implemented followed their papers on the platform of Pytorch 1.10.

D. Further Performance Analysis

Results in Table III reveals that traditional anomaly detection algorithms such as OCSVM [13] and IF [14] are not good options for complex anomaly detection for MTS due to their inherent restricted learning ability. DAGMM [20] focuses on anomaly detection for multidimensional data, but in time series anomaly detection, it only analyzes the current point of observation and disregards historical information. However, it is essential to

TABLE VII
PERFORMANCE SIGNIFICANCE ANALYSIS EXPERIMENT

	SWaT	WADI	SMD	SMAP	MSL
OCSVM	***	***	***	***	***
IF	***	***	***	***	***
DAGMM	***	***	=	=	*
LSTM-VAE	***	***	=	=	=
BeatGAN	***	***	*	***	=
MTAD-GAT	***	***	=	=	=
USAD	***	***	=	***	**
GDN	***	***	*	**	**
***/=**	**8/0**	**8/0**	**4/4**	**5/3**	**5/3**

consider temporal dependency in the time series research. The LSTM-VAE [22] model combines LSTM and VAE, but it only considers the current moment z_t in the distribution of latent variables, which limits the decoder's ability in the reconstruction of the data and consequently affects anomaly detection performance. BeatGAN [18] and USAD [7] are based on the adversarial training in GAN model, but their performance in anomaly detection is unreliable due to adversarial training being unstable and prone to pattern collapse. Both the MTAD-GAT [19] and the GDN [21] models are based on graph neural networks, which have recently gained popularity. MTAD-GAT is a hybrid anomaly detection model that combines temporal prediction and data reconstruction. In addition, it integrates prediction error and reconstruction error for anomaly scoring, thereby strengthening the model's robustness. However, the error-based anomaly scoring mechanism is verified inferior to the probability-based one through the ablation experiments.

At the same time, we found that the performance of HybridAD on the SMD, SMAP, and MSL datasets was not significantly different from that of some baseline models. Therefore, in order to give a more objective and comprehensive display of the performance of HybridAD, we used the Mann-Whitney Test [36] (which is a non-parametric statistical test) to verify whether the performance difference between HybridAD and other baseline models on the five datasets is significant. The significance level was set to 0.05. Specifically, we conducted 10 repeated experiments for each model on each dataset and collected the best performance of the model in each experiment as its optimal performance. Table VII shows whether the performance difference between HybridAD and multiple baseline models is significant or not significant on the five datasets. '*' indicates that the performance improvement of HybridAD on that dataset is significant, '=' indicates that the performance improvement of HybridAD on that dataset is not significant. Moreover, we further divided the significant combinations (where p-value $<$ 0.05) into three categories based on the magnitude of the calculated p-values using the Mann-Whitney Test: strongly significant (p-value $<$ 0.001, denoted by '***'), moderately significant (where 0.001 $<$ p-value $<$ 0.01, denoted by '**'), and marginally significant (where 0.01 $<$ p-value $<$ 0.05, denoted by '*').

We can see that HybridAD has significant performance improvement over other baseline models on the SWaT and WADI datasets, while on the other three datasets, models such as LSTM-VAE and MTAD-GAT also exhibit comparable performance. This phenomenon is related to the distribution of data in the dataset and the complexity of the anomaly patterns. At the same time, we can see that deep learning-based anomaly detection models have greatly improved performance compared to traditional anomaly detection models (such as IF and OCSVM), which means that deep learning models still have great research value in anomaly detection tasks.

E. Details of the Ablation Experiment

In Section IV-C2, we designed five model variants to verify the effectiveness of the probability density model designed in HybridAD. Each variant corresponds to an anomaly score (which is strongly correlated with the model design). The first three value-based model variants (ED-Only Prediction, ED-Only Reconstruct, ED-Composite) typically calculate anomaly scores as the Euclidean distance between the model's output and the actual observation values. The latter two probability density-based model variants (PD-Only Prediction, PD-Only Reconstruct) calculate anomaly scores as the negative logarithm of the probability density of the actual observation values under the distribution output by the model. The specific definitions of the anomaly scores in each model variant are shown below.

ED-Only Prediction: The anomaly score takes into account only the prediction error, which is the mean square error between predicted and actual observations for each dimension, as demonstrated by (12).

$$S_{x_{t+1}} = \frac{1}{M} \sum_{i=1}^{M} \left(\hat{x}_{t+1}^{i} - x_{t+1}^{i}\right)^2, \tag{12}$$

where M is the dimension size of the sequence.

ED-Only Reconstruct: The anomaly score only takes reconstruction error into account. As shown in (13), the error of the observation at the moment t in the reconstruction sequence is utilized as the anomaly score in this work.

$$S_{x_t} = \frac{1}{M} \sum_{i=1}^{M} \left(\hat{x}_{t}^{i} - x_{t}^{i}\right)^2. \tag{13}$$

ED-Composite: This anomaly score takes into account the reliability of the prediction source, which is calculated as the sum of the weighted reconstruction error and the prediction error of the sequence, as shown in (14).

$$S_{x_{t+1}} = \frac{\sum_{i=t-L+1}^{t} d_i \times \sum_{j=1}^{M} \left(\hat{x}_{i}^{j} - x_{i}^{j}\right)^2}{D_L} + \frac{1}{M} \sum_{i=1}^{M} \left(\hat{x}_{t+1}^{i} - x_{t+1}^{i}\right)^2, \tag{14}$$

where $d_i = \alpha^{i-(t-L+1)}$ and $D_L = \sum_{i=t-L+1}^{t} d_i$. In experiment, α is a constant set to 1.25.

PD-Only Prediction: The anomaly score considers only the prediction probability density, as specified in (15).

$$S_{x_{t+1}} = -\log p\left(x_{t+1} \mid \mu_{x_{t+1}}, \sigma_{x_{t+1}}\right). \tag{15}$$

PD-Only Reconstruction: The anomaly score considers only the reconstruction probability density and is defined as the reconstruction probability density of the observation x_t in the reconstructed sequence W_t, as given in (16).

$$S_{x_{t+1}} = -\log p\left(x_t \mid \mu_{x_t}, \sigma_{x_t}\right). \tag{16}$$

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